Dynamical Processes in complex networks

Requirements

- Newtonian Mechanics
- Basics of Matrices/Linear Algebra
- Matlab/Python/C++
- Graphical Plot (Must)
- Differential Equations
- Computational Complexity
- Error calculation

Instructor: Chittaranjan Hens
CCNSB

Dynamical Processes in complex networks

Requirements

- Newtonian Mechanics
- Basics of Matrices/Linear Algebra
- Statistical Mechanics
- Matlab/Python/C++
- Graphical Plot (Must)
- Differential Equations
- Computational Complexity
- Error calculation

- "**Ludwig Boltzmann**, who spent much of his life Studying Statistical Mechanics, died in 1906 by his own hand".
- "His student, **Paul Ehrenfest**, carrying on **Boltzmann**'s work, died similarly in 1933".
- ***"Now it is our turn to study statistical mechanics. Perhaps it will be wise to approach the subject cautiously".***

- From the book **"States of Matter"** by David L. Goldstein (Dover, 1985)

Instructor: **Chittaranjan Hens**
CCNSB

Module 1 Introduction
Complex systems

Statistical physics and interacting particles

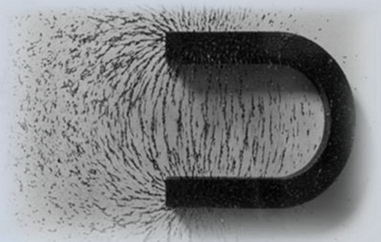
microscopic units

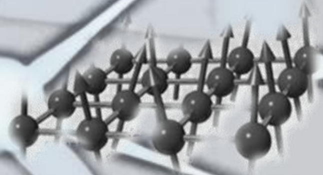
macroscopic observations

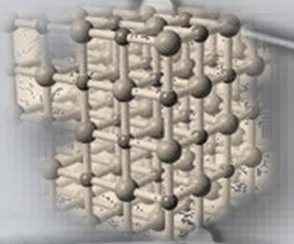
connectivity map

$$\mathcal{H} = -\sum_{i,j} \mathcal{J}_{ij}\sigma_i\sigma_j$$

Ferromagnetism





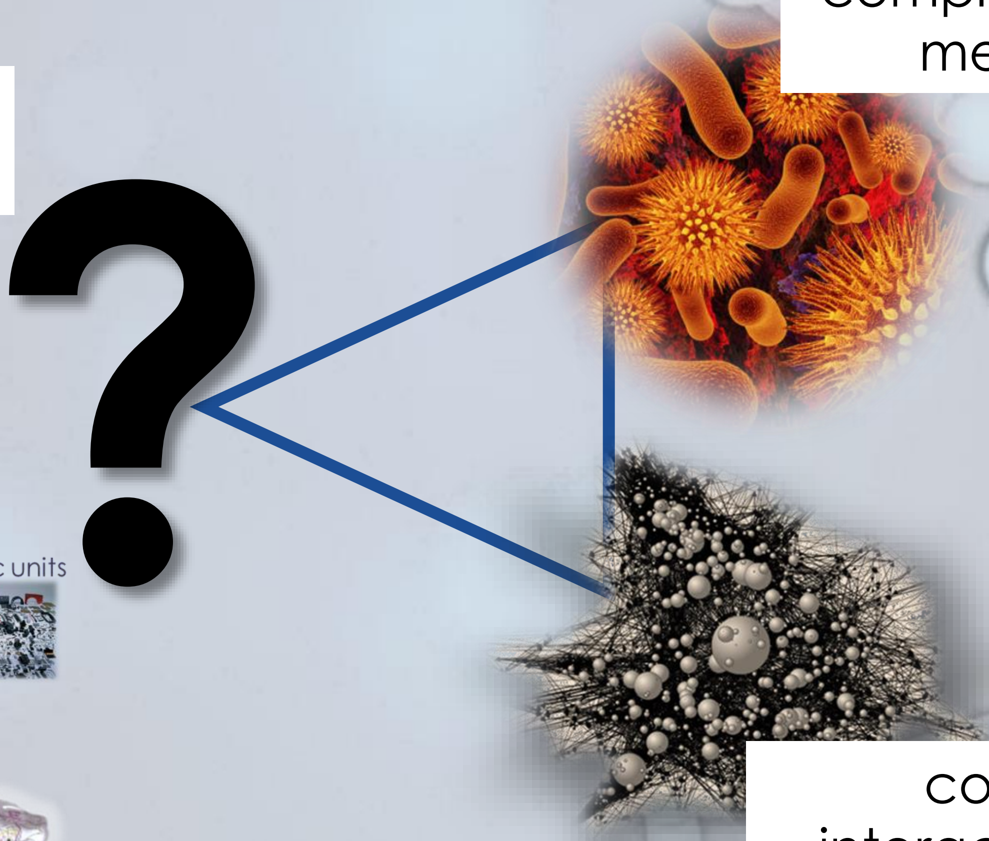


linking macroscopic behavior to microscopic models

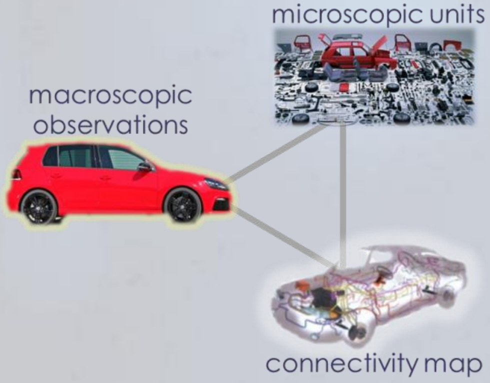
Statistical physics and interacting particles

macroscopic observation

complex interaction mechanisms

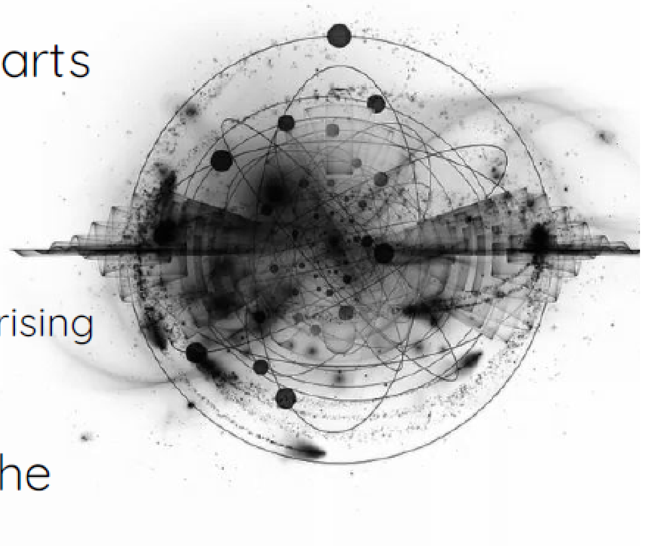


complex interaction maps



Complex Systems

- consists of many interconnected parts
- characterized by time-dependent interactions among their parts
- are not an aggregation of their separate parts
- when looked at as a whole gives non trivial insights
    - ***Emergence***: a property not any of components have on their own, arising during self-organization process
- Often interactions change states of parts, and the states of the parts change the networks of interactions



com·plex

adjective
/ˌkämˈpleks,kəmˈpleks,ˈkämˌpleks/

1. consisting of many different and connected parts.
"a complex network of water channels"
synonyms: compound, composite, compounded, multiplex
"a complex structure"

Complex Systems

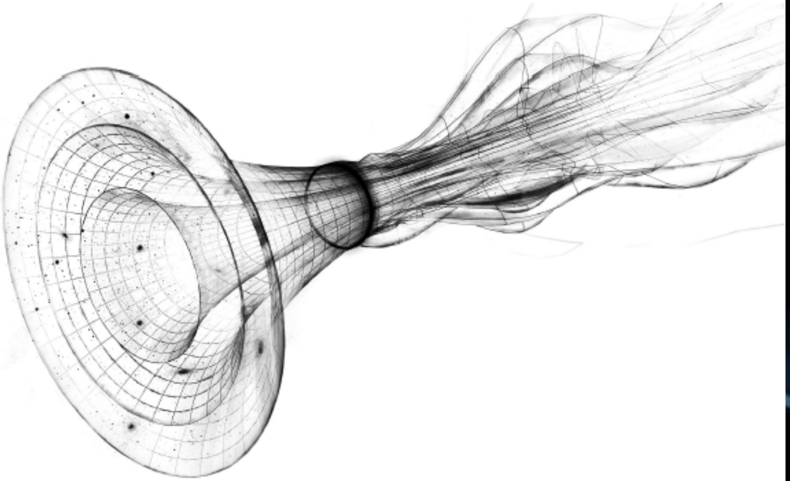
Physics

Examples: deterministic chaos, quantum entanglement, spin glasses

Study of complex systems has a long history in Physics, dating back to Aristotle's time, and more relevant than ever in this century

"I think the next [21st] century will be the century of complexity"

— Stephen Hawking



More is different,

— P. Anderson, Science (1972)

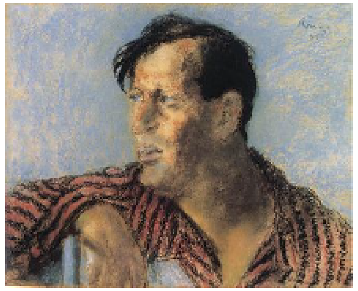
philosophy of science and emergent phenomena; limitations of reductionism and the existence of hierarchical levels of science

Module 1 Introduction

Why Network Science?


THE EXPONENTIAL GROWTH OF HUMANITY
Now ~7.8 Billion
1800
1 BILLION
1930
2 BILLION
1960
3 BILLION
1974
4 BILLION
1987
5 BILLION
1999
6 BILLION
2011
7 BILLION

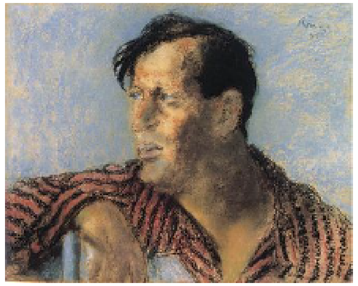
Why Network Science?



1929 Frigyes Karinthy

"If you choose a person out of the 1.5 billions of our planet, I bet that using no more than ***five*** individuals, one of them my acquaintance, I could contact the person you chose, using only the list of acquaintances of each one"

Why Network Science?



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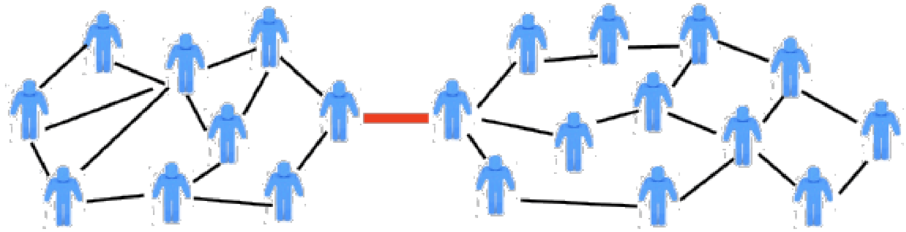


Why Network Science?

Complex networks– Construction 1

- In 1967, psychologist Stanley Milgram coined the phrase "*six degrees of separation*" to describe the world-shrinking effects of social networks.

Why Network Science?

Complex networks– Construction 1

1969 Stanley Milgram

- People chosen at random on a US State
- Request to send a letter to a given final person in another state :
  - If you know the final person, send directly to him
  - If not, send to someone you think it is more likely to know him

An Experimental Study of the Small World Problem*

JEFFREY TRAVERS
Harvard University
AND
STANLEY MILGRAM
The City University of New York

Arbitrarily selected individuals (N=296) in Nebraska and Boston are asked to generate acquaintance chains to a target person in Massachusetts, employing "the small world method" (Milgram, 1967). Sixty-four chains reach the target person. Within this group the mean number of intermediaries between starters and targets is 5.2. Boston starting chains reach the target person with fewer intermediaries than those starting in Nebraska; subpopulations in the Nebraska group do not differ among themselves. The funneling of chains through sociometric "stars" is noted, with 48 per cent of the chains passing through three persons before reaching the target. Applications of the method to studies of large scale social structure are discussed.

Why Network Science?

Complex networks– Construction 1

1969 Stanley Milgram

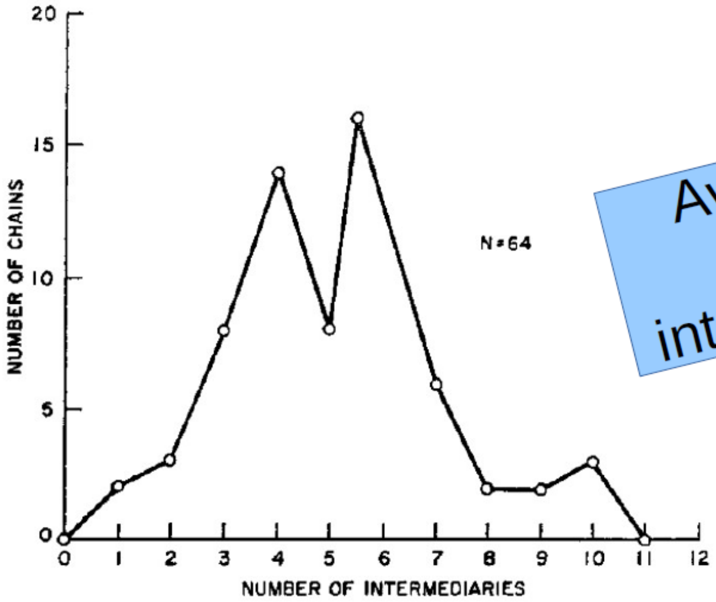
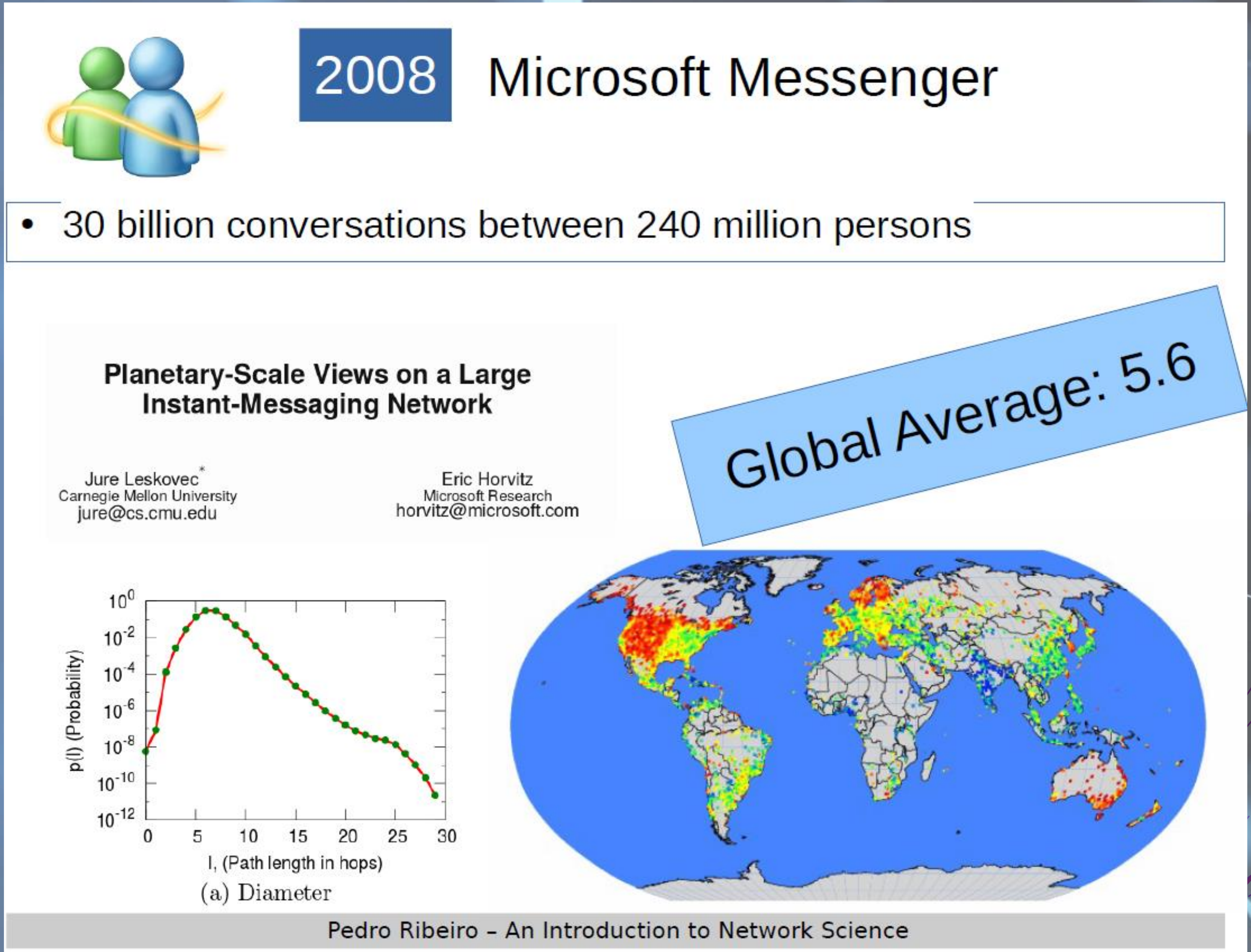


FIGURE 1

Lengths of Completed Chains

Average between 5.5 and 6 intermediate persons

Why Network Science?



Why Network Science?

- Imagine that a person has, on average, 100 friends
  - 0 intermediates: 100
  - 1 intermediate: 100^2 = 10.000
  - 2 intermediates: 100^3 = 1.000.000
  - 3 intermediates: 100^4 = 100.000.000
  - 4 intermediates: 100^5 = 10.000.000.000
  - 5 intermediates: 100^6 = 1.000.000.000.000

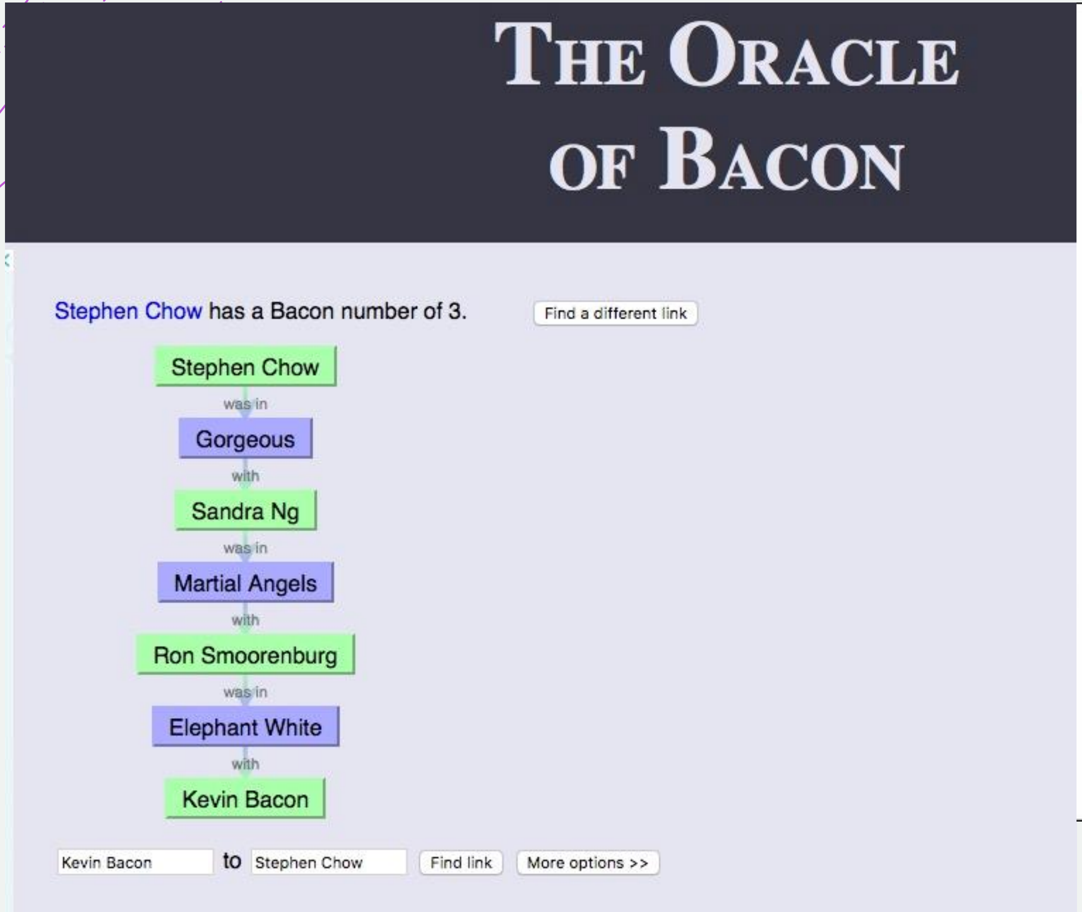
The power of exponentiation

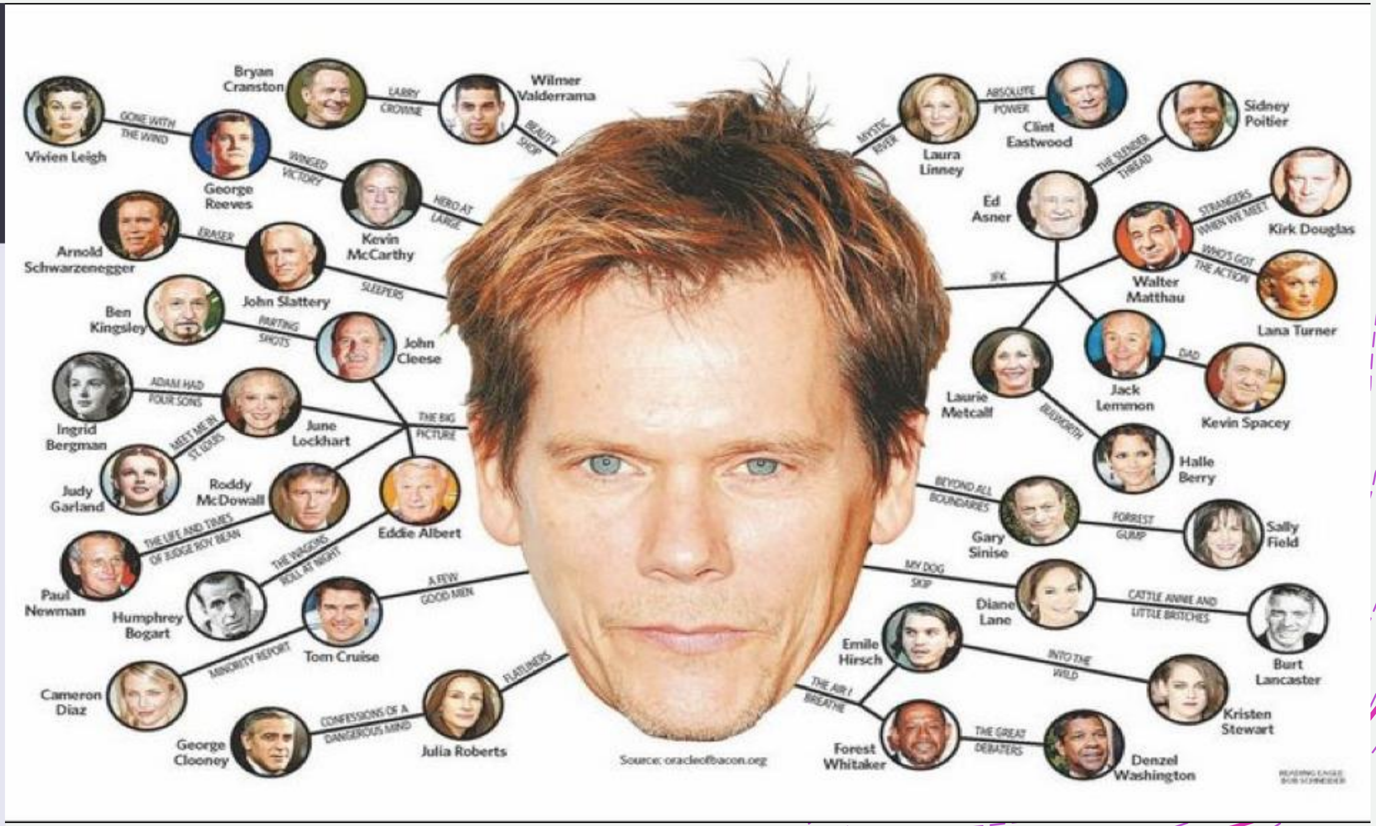
- In practice, not all friends are new, but still there is a very fast growth

Why Network Science?

Complex networks– Construction 2

- *Kevin Bacon Game http://oracleofbacon.org/movielinks.php*

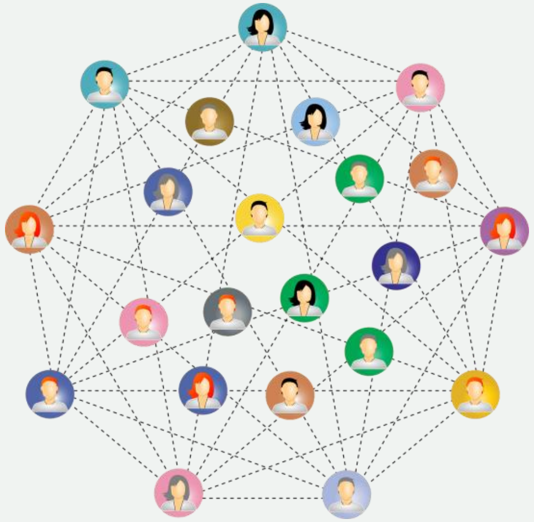


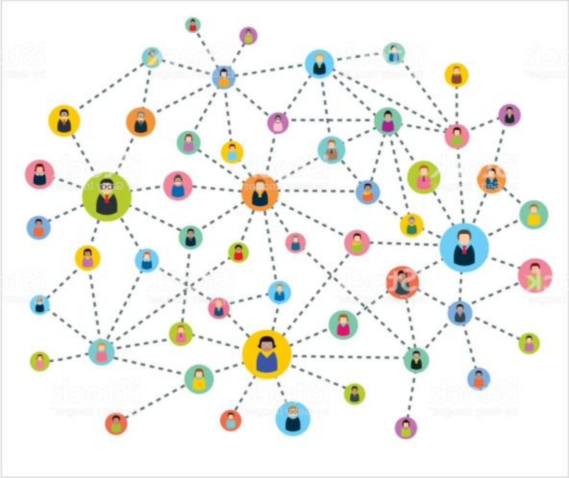


Why Network Science?

Complex networks– Construction 3

+ *Friendship network*



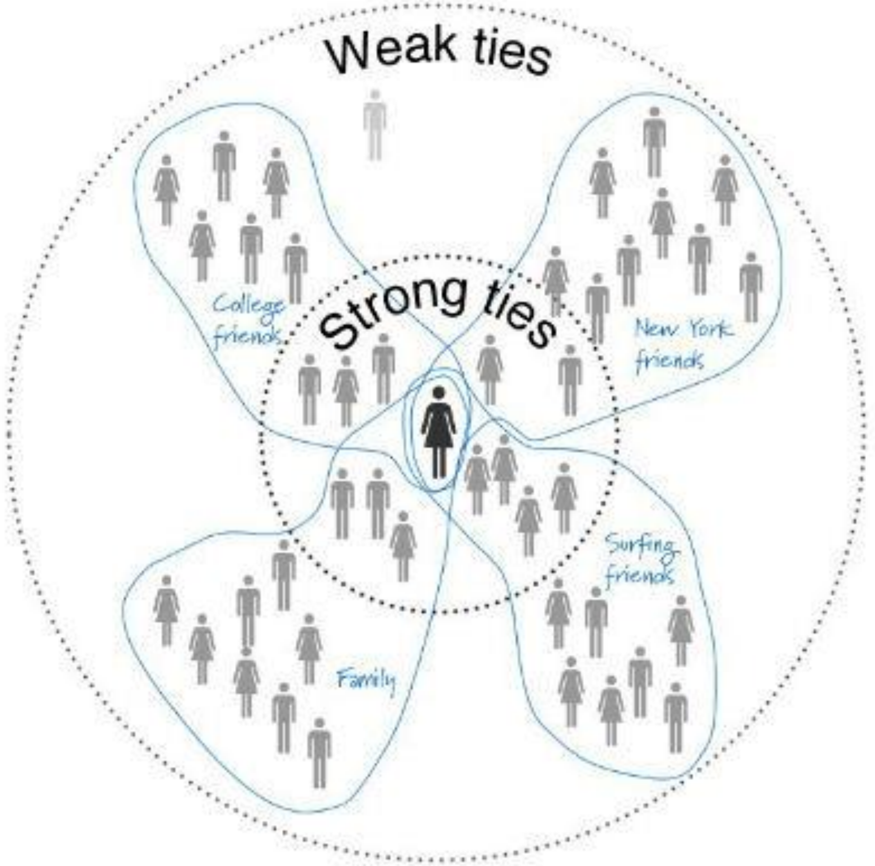


Time evolution

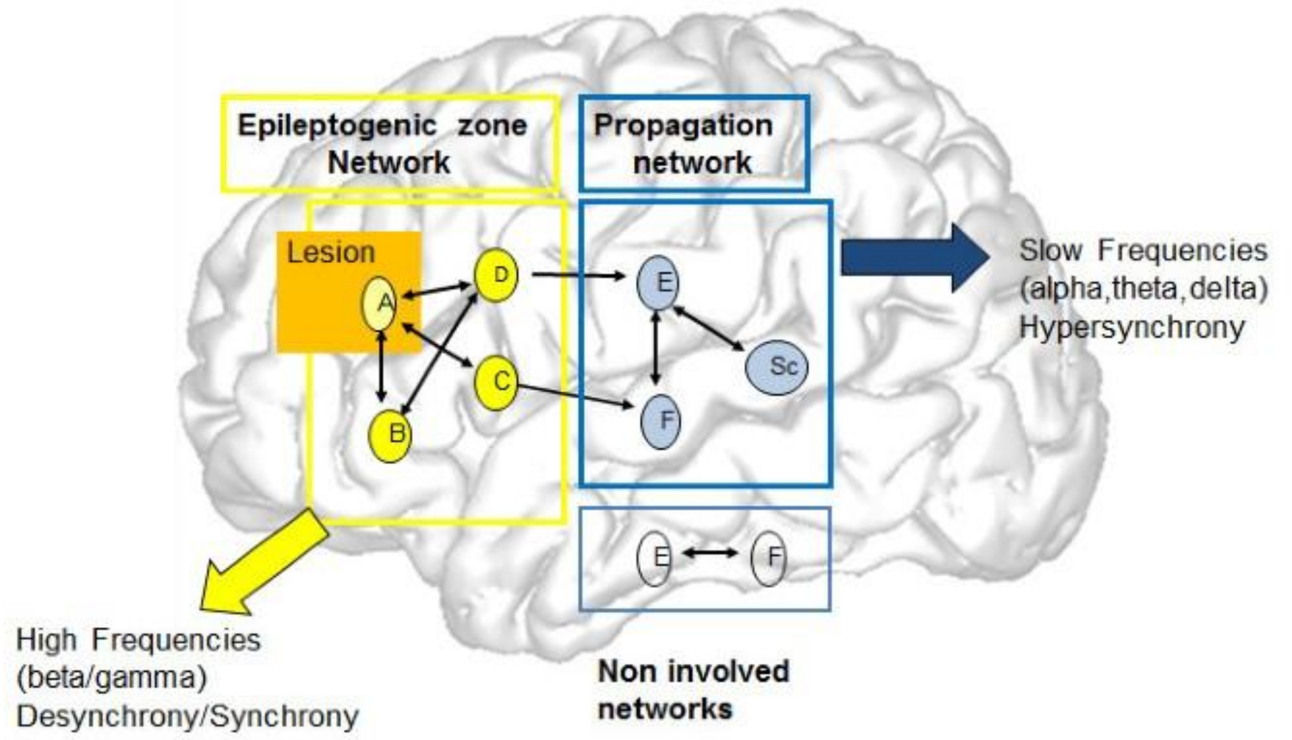
+ *Limits of friendship*

Complex networks– Functionality 1

+ *Weak ties*



Friendship



Brain network of networks, Integration

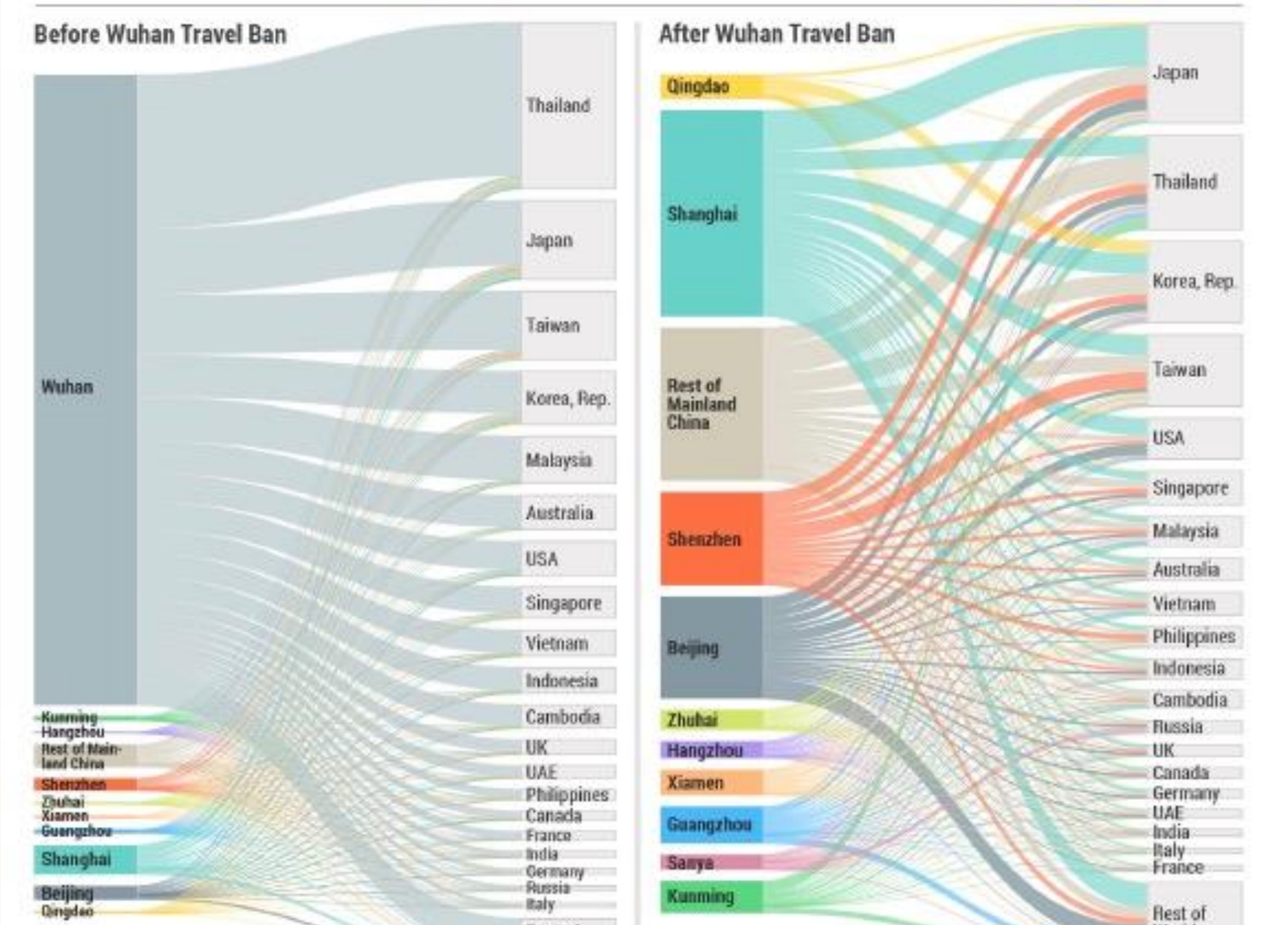
Complex networks– Functionality 2

+ *Power grid stability*

2006, European power grid, cascading

Complex networks– Functionality 3

+ *COVID-19 epidemic spreading*



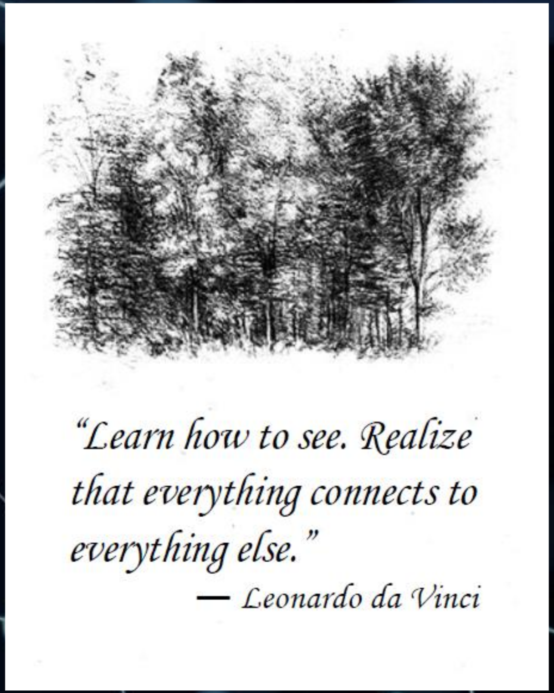
Will return after few slides

Network Science: Interdisciplinary

Complex Networks is a blend of several disciplines

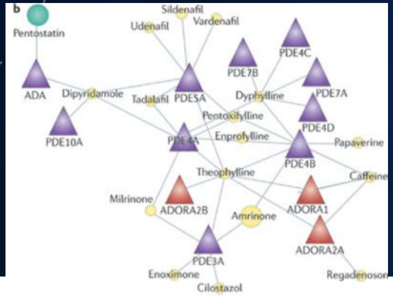
- Mathematics (Random graph theory)
- Statistics (Statistical inference)
- Computer Science (Graph algorithms)
- Physics (Statistical physics, Percolation theory)
- Sociology (Data collection and analysis)

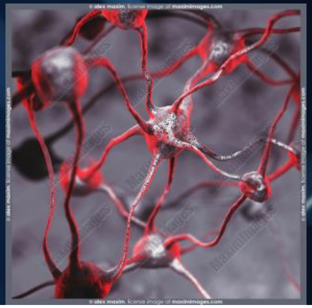
Connections

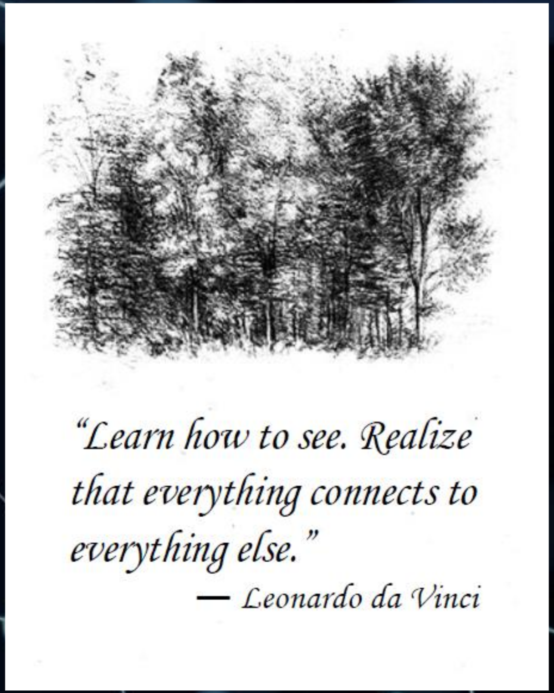


Network Science and Complex Systems

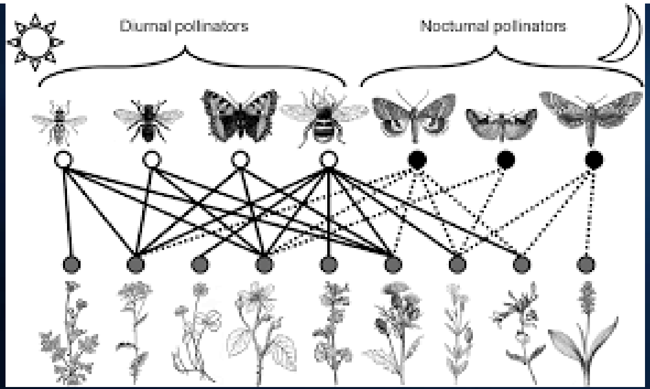
Connections

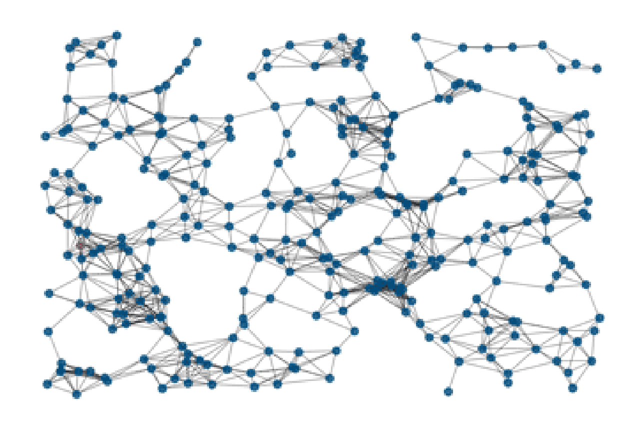






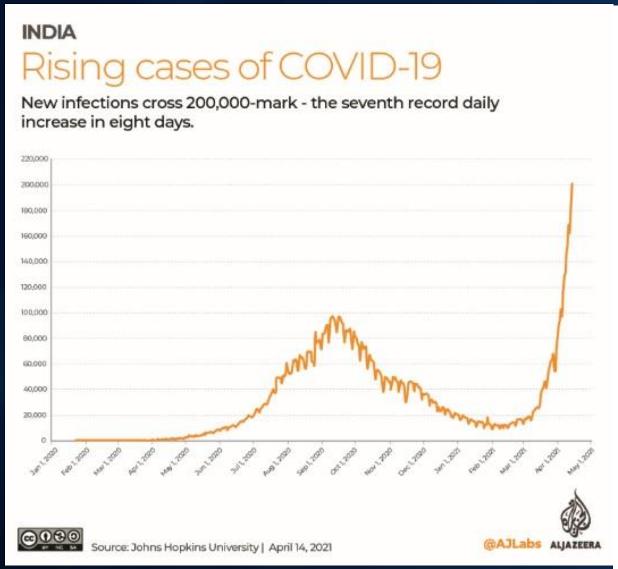
Interactions

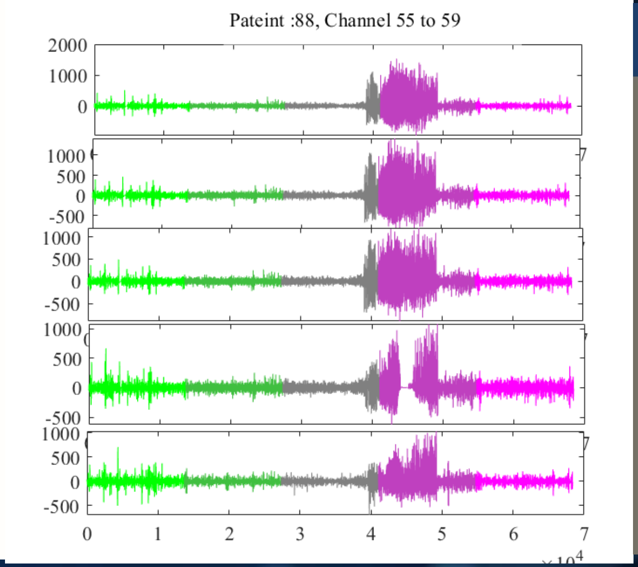


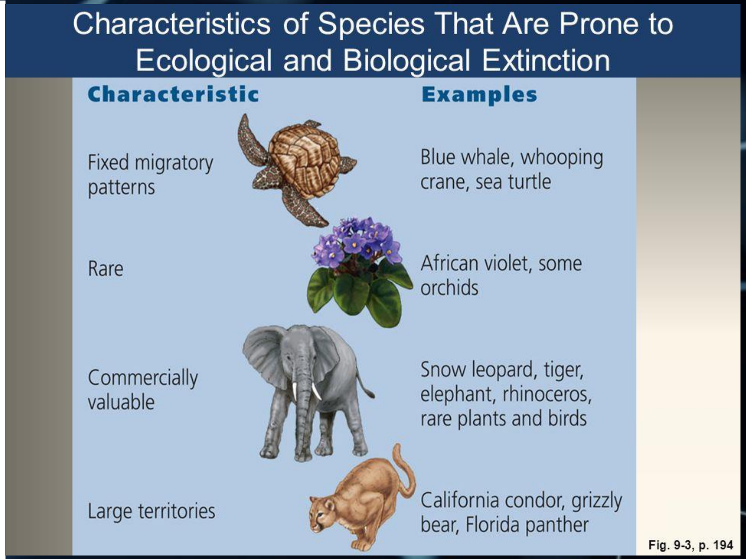


Why Network Science?

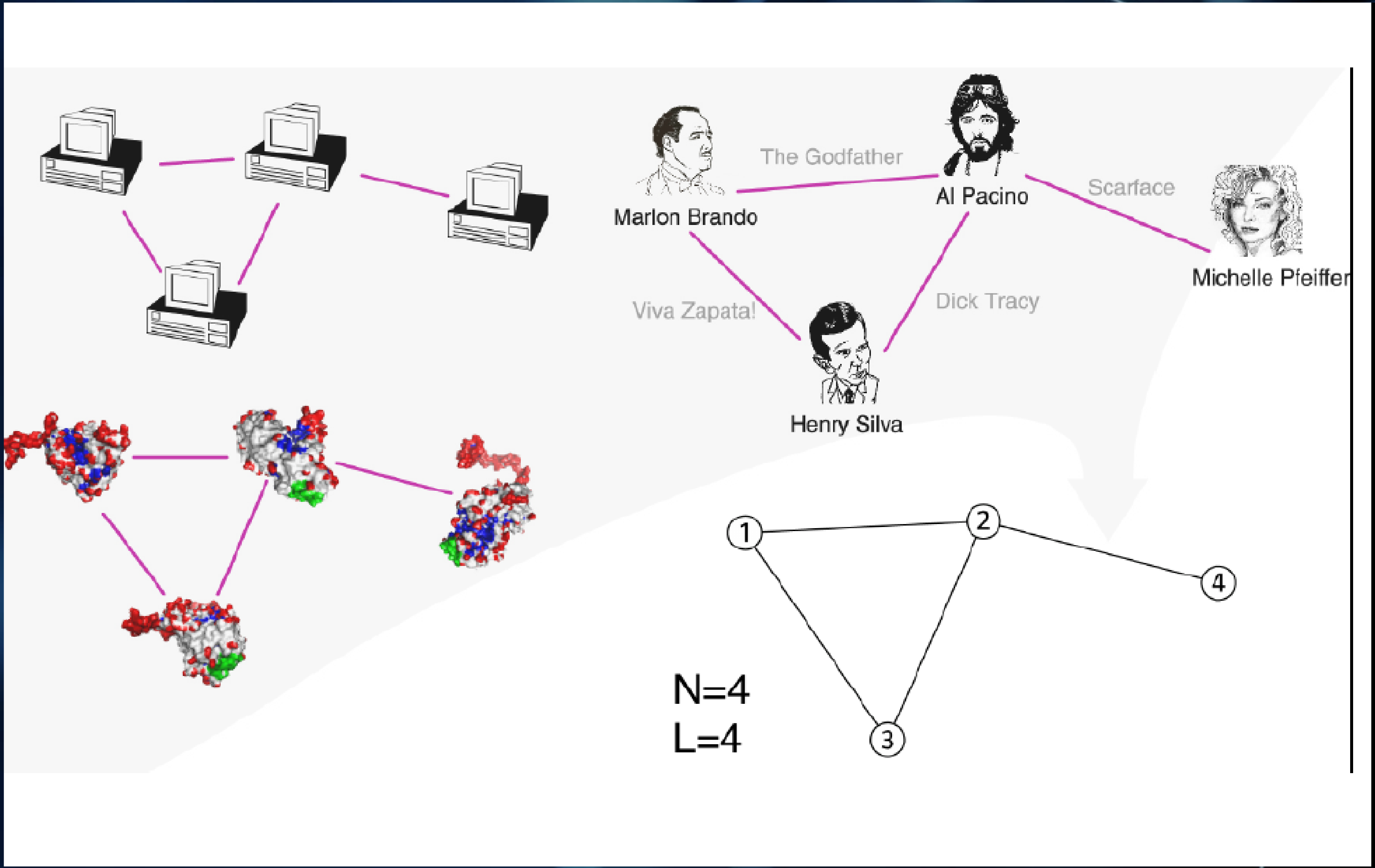
Emergence







Why Network Science?

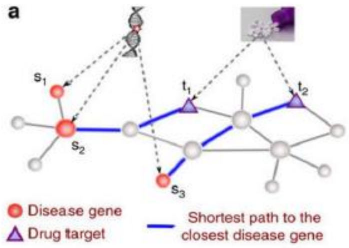


Why Network Science?

Applied sciences

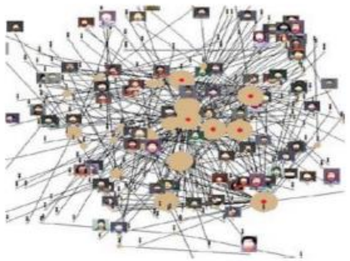
Interconnected systems exist in many applied sciences and other fields. There are numerous studies which show looking at these compex system, as a whole, gives us non trivial insights and is necessary to understand these systems.

Medicine



Disease Gene Network
Credit: Guney et al. (2016)
"the emergence of most diseases cannot be explained by single-gene defects, but involve the breakdown of the coordinated function of distinct gene groups"

Law

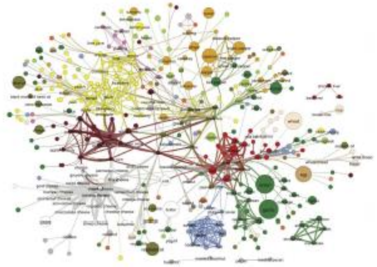


Criminal Network
Credit: Xu et al. (2005)

Economics

Trading Network
Credit: Adamic et al. (2017)
"strong feedback between the trading behaviour in buyers and sellers networks and the market conditions"

Culinary



Flavor Network
Credit: Ahn et al. (2011)

Read on food pairing theories and check out the interactive demo: https://foodgalaxy.jp/

Why Network Science?

Networks: Nodes and Links

NETWORK	NODES	LINKS	DIRECTED UNDIRECTED	N	L
Internet	Routers	Internet connections	Undirected	192,244	609,066
WWW	Webpages	Links	Directed	325,729	1,497,134
Power Grid	Power plants, transformers	Cables	Undirected	4,941	6,594
Mobile Phone Calls	Subscribers	Calls	Directed	36,595	91,826
Email	Email addresses	Emails	Directed	57,194	103,731
Science Collaboration	Scientists	Co-authorship	Undirected	23,133	93,439
Actor Network	Actors	Co-acting	Undirected	702,388	29,397,908
Citation Network	Paper	Citations	Directed	449,673	4,689,479
E. Coli Metabolism	Metabolites	Chemical reactions	Directed	1,039	5,802
Protein Interactions	Proteins	Binding interactions	Undirected	2,018	2,930

Why Network Science?

Dissecting complex systems/connectivity using network

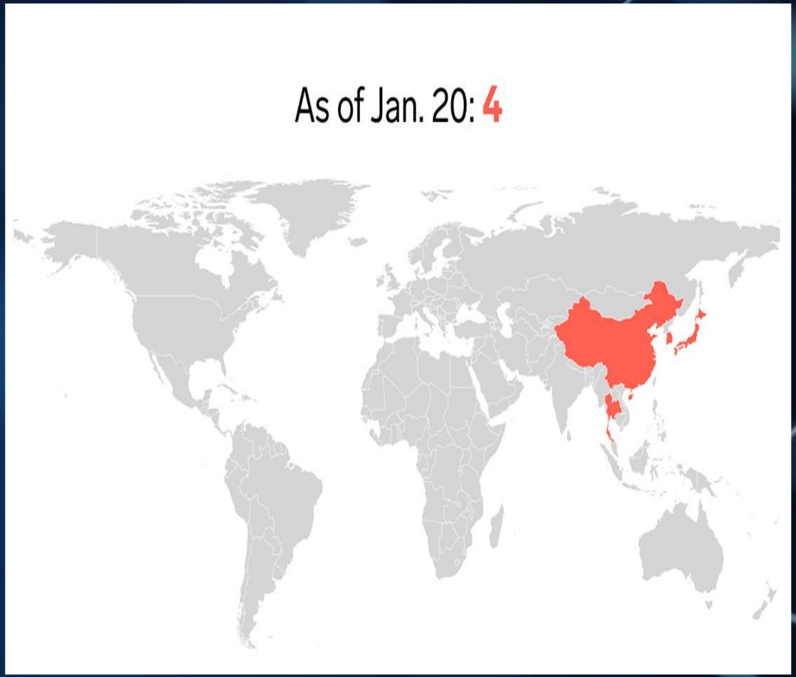
H1N1 pandemic

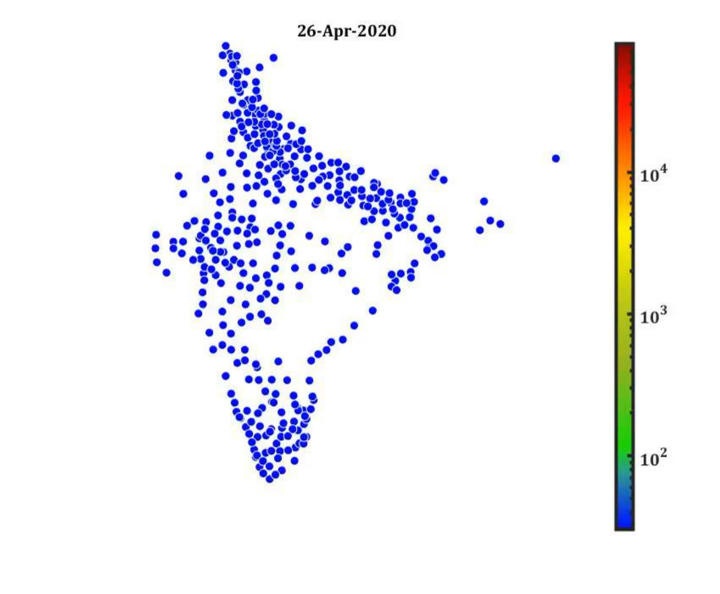
http://networksciencebook.com/chapter/1#societal-impact

https://www.gleamviz.org/

Video courtesy of Alessandro Vespignani.

Epidemic Spreading





Epidemic Spreading and beyond

- Today epidemic prediction is one of the most active applications of network science, being used to foresee the spread of influenza or to contain Ebola or to understand COVID.

It can be used to model and predict the spread of biological, digital and social viruses (memes).
The impact of these advances are felt beyond epidemiology!!!

Indeed, in January 2010 network science tools have predicted the conditions necessary for the emergence of viruses spreading through mobile phones.

D. Balcan, H. Hu, B. Goncalves, P. Bajardi, C. Poletto, J. J. Ramasco, D. Paolotti, N. Perra, M. Tizzoni, W. Van den Broeck, V. Colizza, and A. Vespignani. Seasonal transmission potential and activity peaks of the new influenza A(H1N1): a Monte Carlo likelihood analysis based on human mobility. BMC Medicine, 7: 45, 2009.

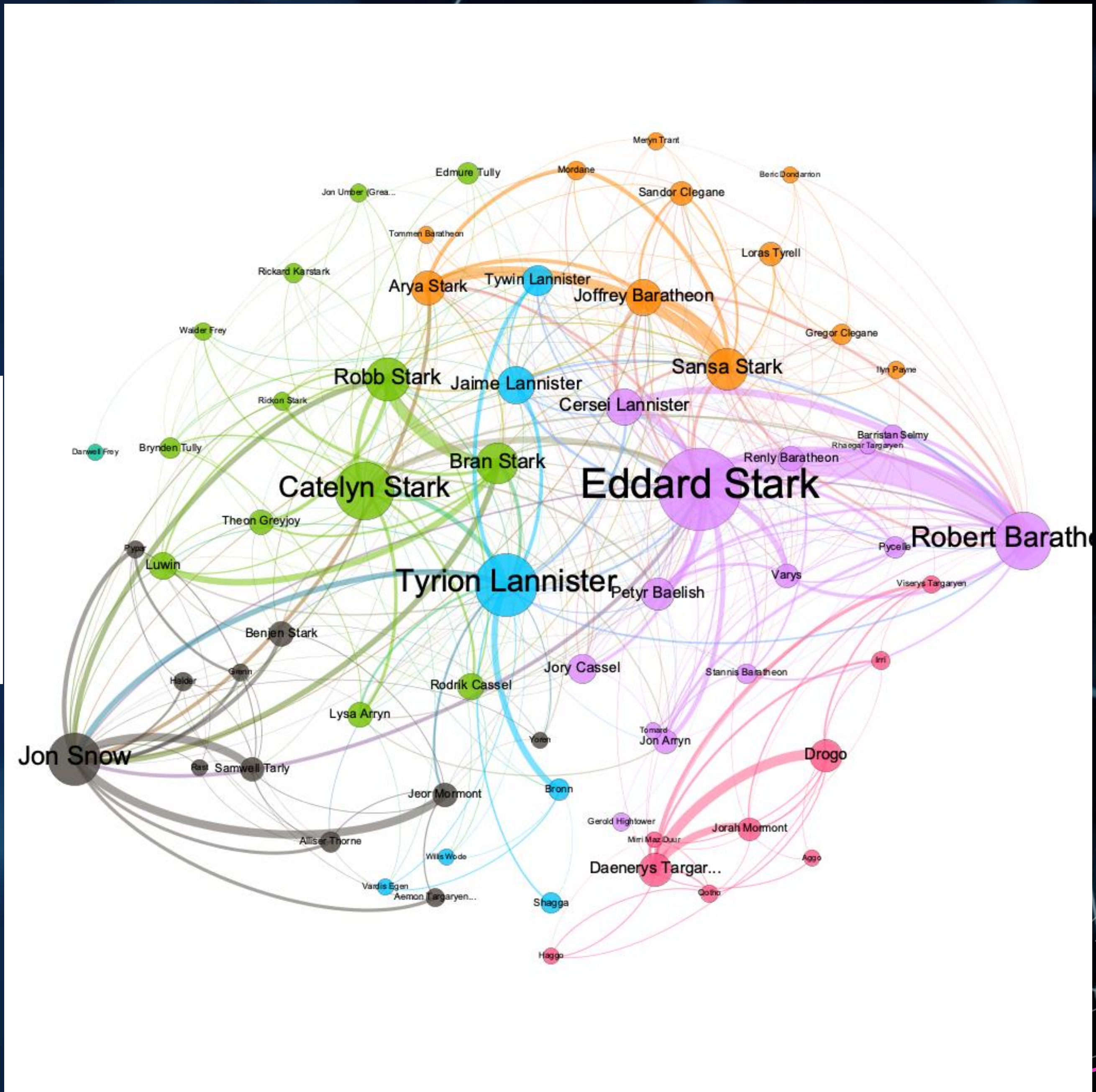
L. Hufnagel, D. Brockmann, and T. Geisel. Forecast and control of epidemics in a globalized world. PNAS, 101: 15124, 2004.

P. Wang, M. Gonzalez, C. A. Hidalgo, and A.-L. Barabási. Understanding the spreading patterns of mobile phone viruses. Science, 324: 1071, 2009.

Epidemic Spreading and beyond
Text analysis

A Song of Ice and Fire by George R.R. Martin.

This is an interaction network and were created by connecting two characters whenever their names (or nicknames) appeared within 15 words of one another in one of the books. The edge weight corresponds to the number of interactions.

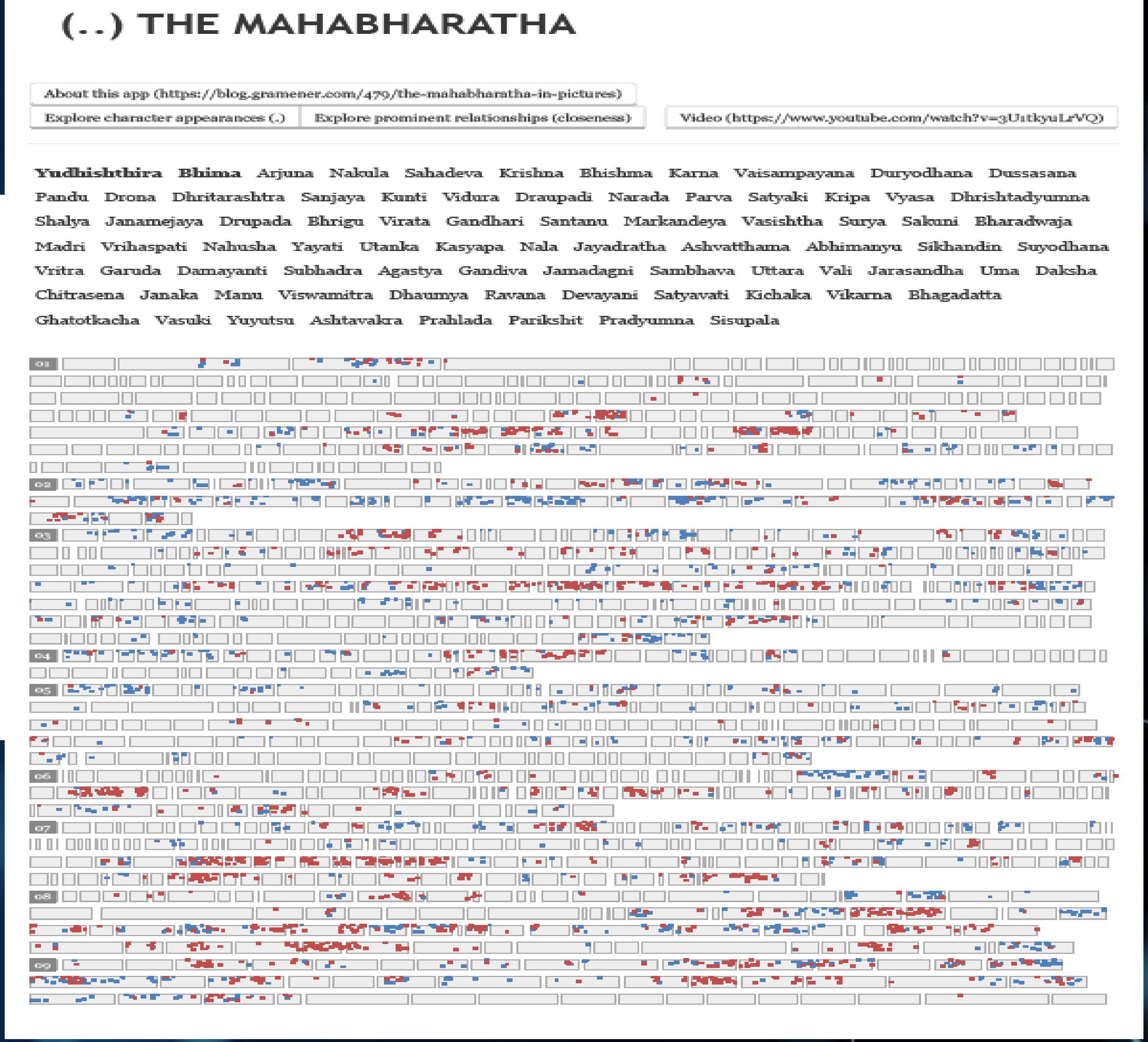


https://ericmjl.github.io/Network-Analysis-Made-Simple/05-casestudies/01-gameofthrones/

Epidemic Spreading and beyond

Text analysis

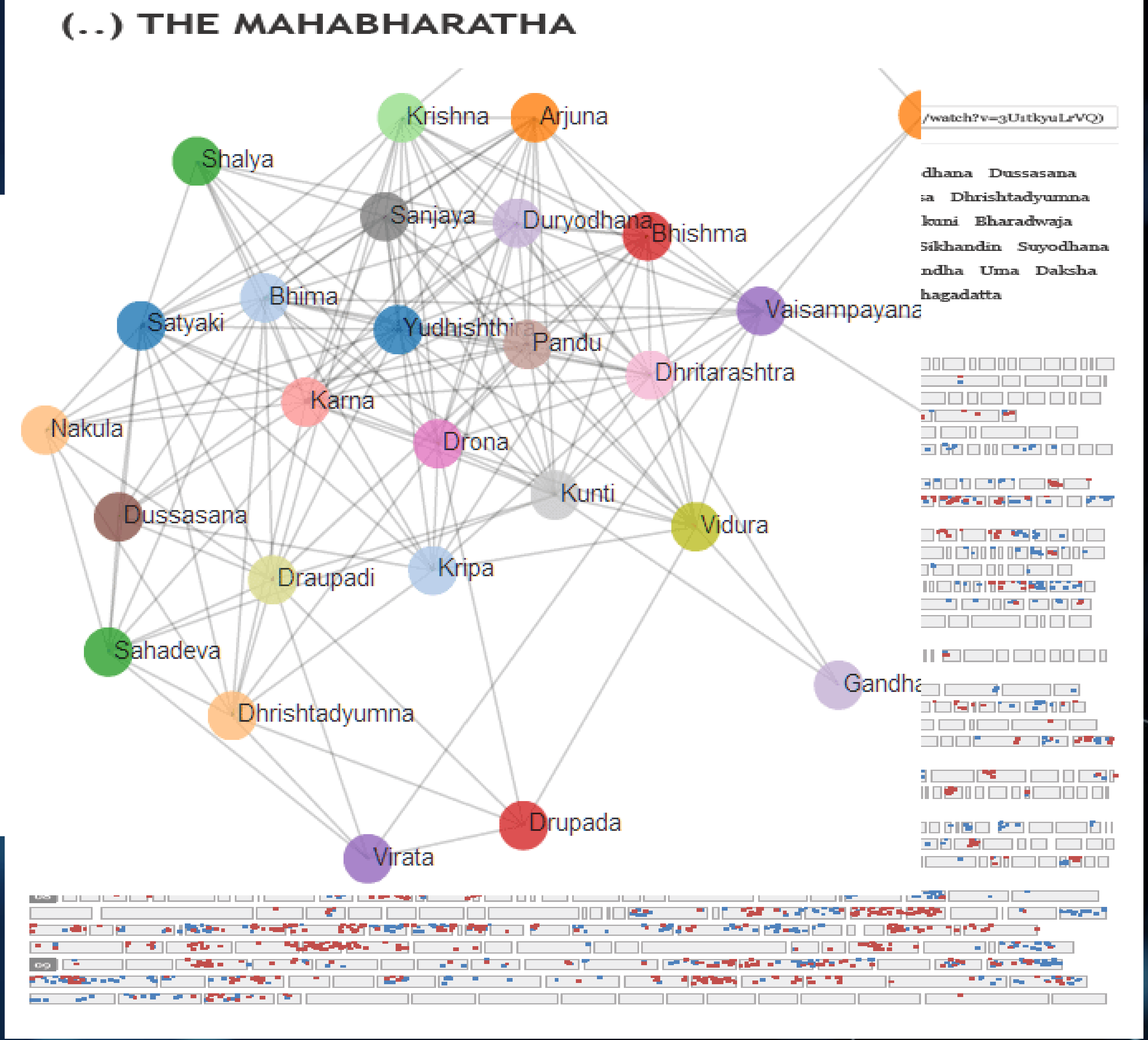
- 18 Parvas
- Total 98 Subparvas
- Verses 200,000 ("the longest poem ever written")
- At about 1.8 million words in total
- Roughly ten times the length of the Iliad and the Odyssey combined, or about four times the length of the Ramayana.
- (400 BCE-400 CE)



ttps://blog.gramener.com/the-mahabharatha-in-pictures/

Epidemic Spreading and beyond
Text analysis

- which characters are most closely related? (how often are two characters found within a few words of each other).
- Who are the centrals?
- Who is most responsible for the war?
- Which link or node has high strength?



ttps://blog.gramener.com/the-mahabharatha-in-pictures/

Is network approach really interesting???

Is network approach really interesting???

Let's check some data and statistics

Albert-László Barabási

Northeastern University
, Harvard Medical School

network science
statistical physics
biological physics
physics
medicine

	All
Citations	270320
h-index	159
i10-index	367

4 articles 105

not available a

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TITLE	CITED BY	YEAR
Emergence of scaling in random networks AL Barabási, R Albert Science 286 (5439), 509-512	43097	1999
Statistical mechanics of complex networks R Albert, AL Barabasi Reviews of Modern Physics 74, 47-97	26187	2002
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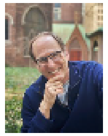

Steven H. Strogatz

Schurman Professor of Applied
Mathematics,
Cornell University

synchronization
complex systems
networks
applied mathematics
nonlinear dynamics

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Citations	123206	42379
h-index	85	56
i10-index	169	125

0 articles 33 articles

not available available

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TITLE	CITED BY	YEAR
Collective dynamics of 'small-world' networks DJ Watts, SH Strogatz Nature 393, 440-442	50657	1998
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Random graphs with arbitrary degree distributions and their applications MEJ Newman, SH Strogatz, DJ Watts Physical review E 64 (2), 026118	4605	2001
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Synchronization of Lorenz-based chaotic circuits with applications to communications KM Cuomo, AV Oppenheim, SH Strogatz IEEE Transactions on circuits and systems II: Analog and digital signal …	1625	1993
Chimera states for coupled oscillators DM Abrams, SH Strogatz	1379	2004

Paul Erdős

Mathematics
number theory
combinatorics
probability
set theory
mathematical analysis

	All	Since 201
Citations	98940	2917
h-index	128	7
i10-index	818	3

TITLE	CITED BY	YEA
On the evolution of random graphs P Erdos, A Rényi Bull. Inst. Internat. Statist 38 (4), 343-347	23289 *	196
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Olaf Sporns

Distinguished Professor, Psychological and Brain Sciences,
Indiana University

Network Science
Computational Neuroscience
Theoretical Neuroscience
Connectome
Connectomics

Citations
h-index
i10-index

5 articles

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Alessandro Vespignani

Sternberg Family Distinguished Professor, Northeastern University, Boston

Network Science
Computational Epidemiology
Data Science
Statistical Physics
Complex Systems

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Citations	85039	43566
h-index	121	95
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Joseph Loscalzo

Professor of Medicine, Harvard Medical School

Vascular Biology
Nitric Oxide
Redox Biology
Systems Biology
Network Science

	All	Since 2018
Citations	126656	37558
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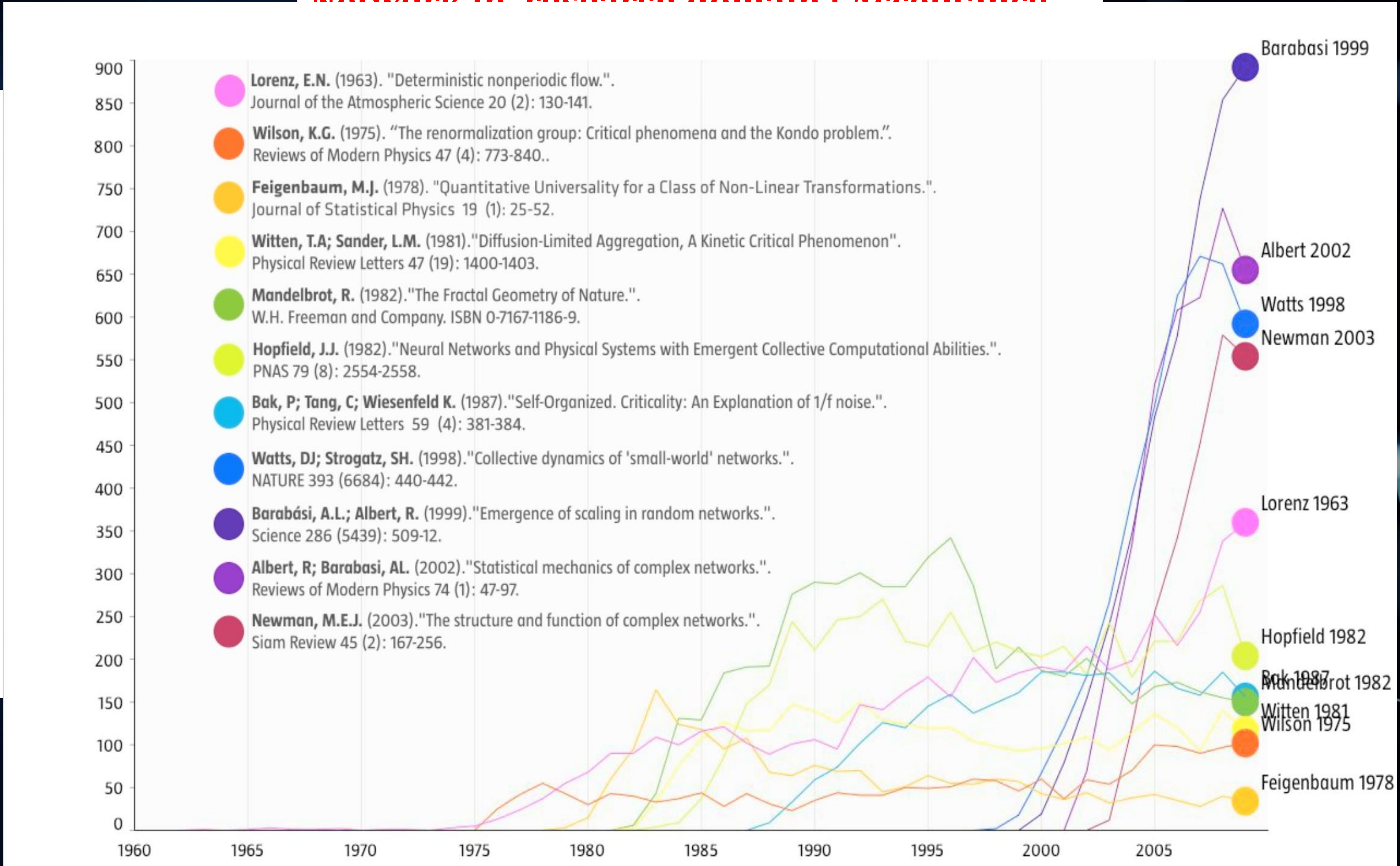
Network in research domain : Acceptance



Network in research domain : Acceptance

a.The 1998 paper by Watts and Strogatz in *Nature* on small world phenomena and the 1999 paper by Barabási and Albert in Science on scale-free networks were identified by Thompson-Reuters as being among the top ten most cited papers in physical sciences during the decade after their publication. Currently (2011) the Watts-Strogatz paper is the second most cited of all papers published in Nature in 1998 and the Barabási-Albert paper is the most cited paper among all papers published in *Science* in 1999.

b.Four years after its publication the *SIAM* review by Mark Newman on network science became the most cited paper of any journal published by the Society of Industrial & Applied Mathematics.

c.*Reviews of Modern Physics*, published since 1929, is the physics journal with the highest impact factor. Until 2012 the most cited paper of the journal was written by Nobel Prize winner Subrahmanyan Chandrasekhar, his classic 1944 review entitled *Stochastic Problems in Physics and Astronomy* . During the 70 years since its publication, the paper gathered over 5,000 citations. Yet, in 2012 it was taken over by the first review of network science published in 2001 entitled *Statistical Mechanics of Complex Networks*.

d.The paper reporting the discovery that in scale-free networks the epidemic threshold vanishes, by Pastor-Satorras and Vespignani, is the most cited paper among the papers published in 2001 by *Physical Review Letters*, shared with a paper on quantum computing.

e.The paper by Michelle Girvan and Mark Newman on community discovery in networks is the most cited paper published in 2002 by *Proceedings of the National Academy of Sciences*.

f.The 2004 review entitled *Network Biology* is the second most cited paper in the history of *Nature Reviews Genetics*, the top review journal in genetics.

R. Pastor-Satorras and A. Vespignani. Epidemic spreading in scalefree networks. Physical Review Letters, 86: 3200, 2001.
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Network in research domain : Acceptance

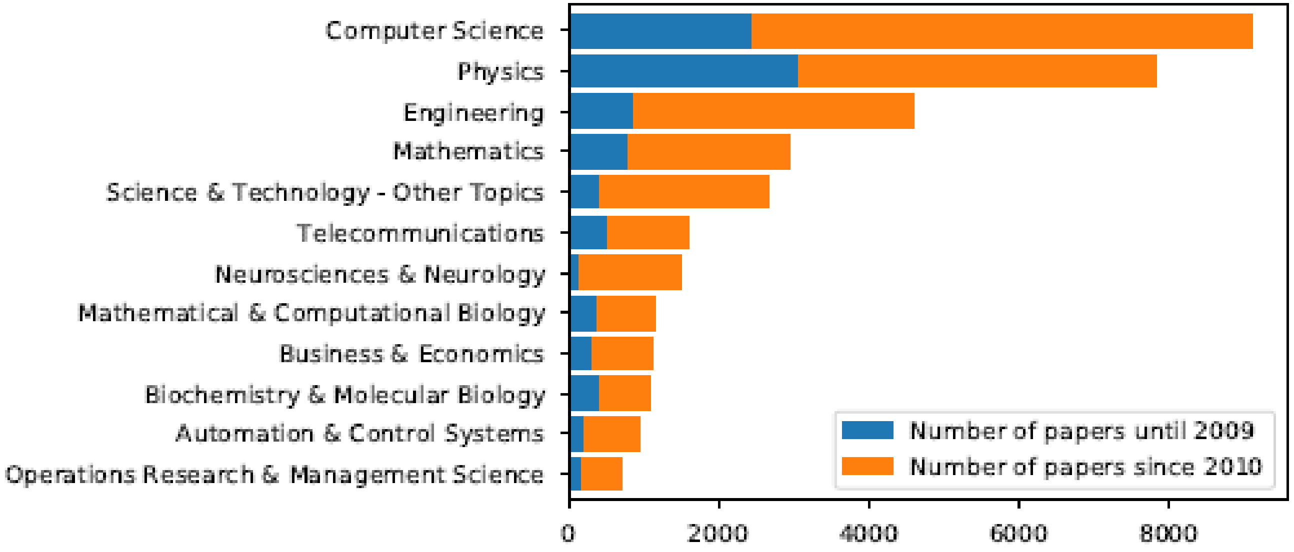


Fig. 3. Top 12 research areas of *network science papers* colored by the decade of publication time.

The Origins of Network Maps

A few of the maps studied today by network scientists were generated with the purpose of studying networks. Most are the byproduct of other projects and morphed into maps only in the hands of network scientists.

a.The list of chemical reactions in a cell were discovered one-byone over a 150 year period by biochemists. In the 1990s they were collected in central databases, offering the first chance to assemble the biochemical networks within a cell.

b.The list of actors that play in each movie were traditionally scattered in newspapers, books and encyclopedias. With the advent of the Internet, these data were assembled into central databases, like imdb.com, feeding the curiosity of movie aficionados. The database allowed network scientists to reconstruct the affiliation network behind Hollywood.

c.The list of authors of millions of research papers were traditionally scattered in the table of content of thousands of journals. Recently Web of Science, Google Scholar, and other services have assembled them into comprehensive databases, allowing network scientists to reconstruct accurate maps of scientific collaboration networks.

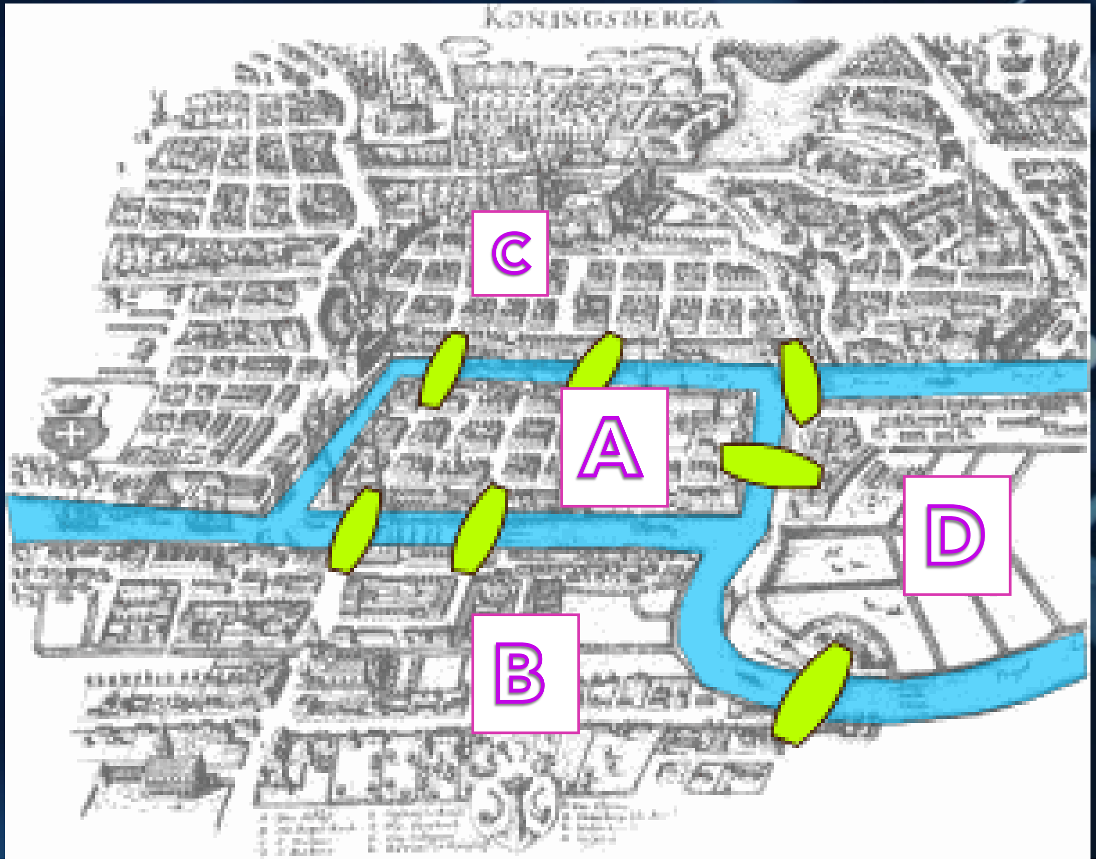
Much of the early history of network science relied on the investigators' ingenuity to recognize and extract networks from preexisting databases. Network science changed that: Today well-funded research collaborations focus on map making, capturing accurate wiring diagrams of biological, communication and social systems.

History of Graph Theory

Origins

- 1735: Euler was puzzled by solving the bridges of Königsberg (origins of graph theory)
- 1950: Erdős–Rényi puzzled by the social networks structure
- 1969: Social Experiment: Milgram
- 1998: Stogratz, Watts: Small world network
- 1999: Barabási, Albert: puzzled by the Internet
- Now: we are puzzled by all of them (brain, social networks, communication and transportation networks, etc)

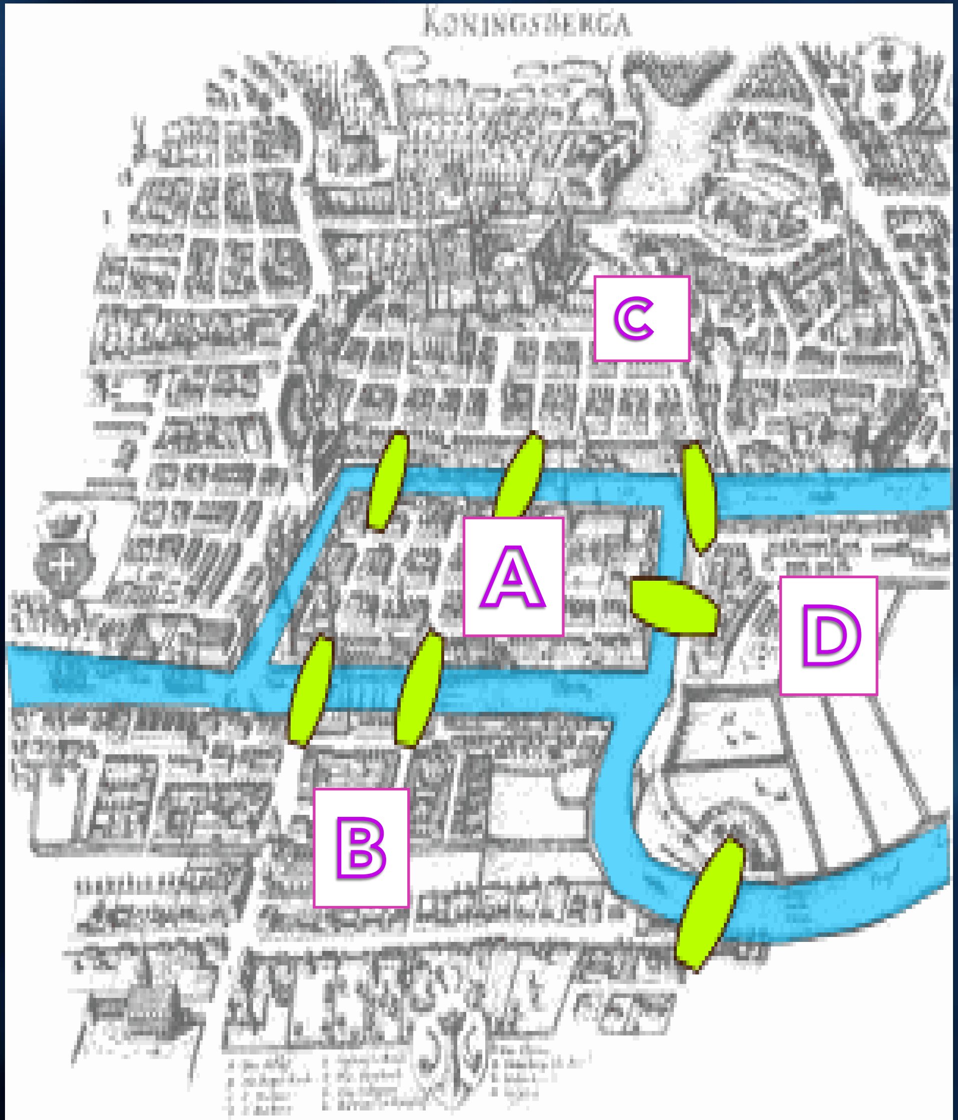
The 7 bridges of Königsberg



The city of Königsberg, Germany, is divided by the Pregal River. The river forks and rejoins itself before forking off again.

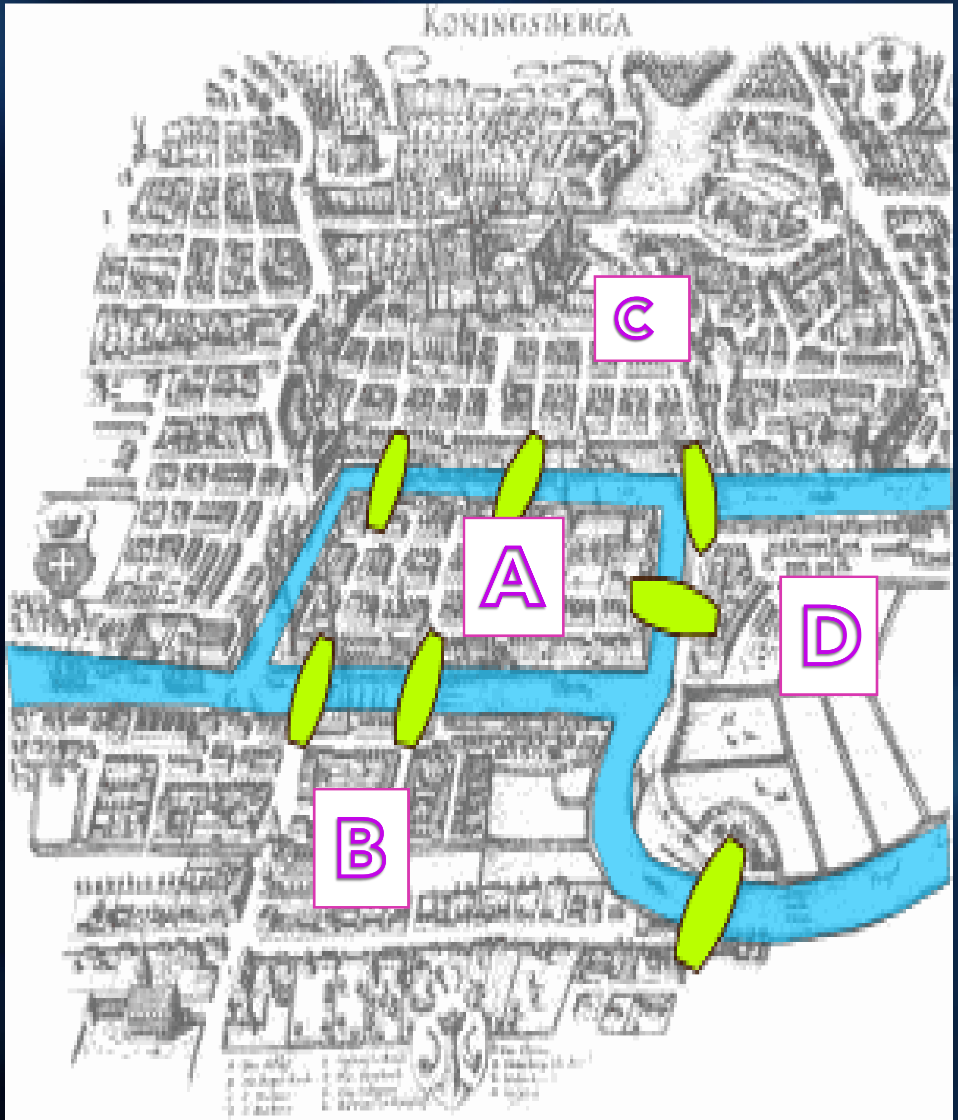
4 land masses and 7 bridges

The 7 bridges of Königsberg



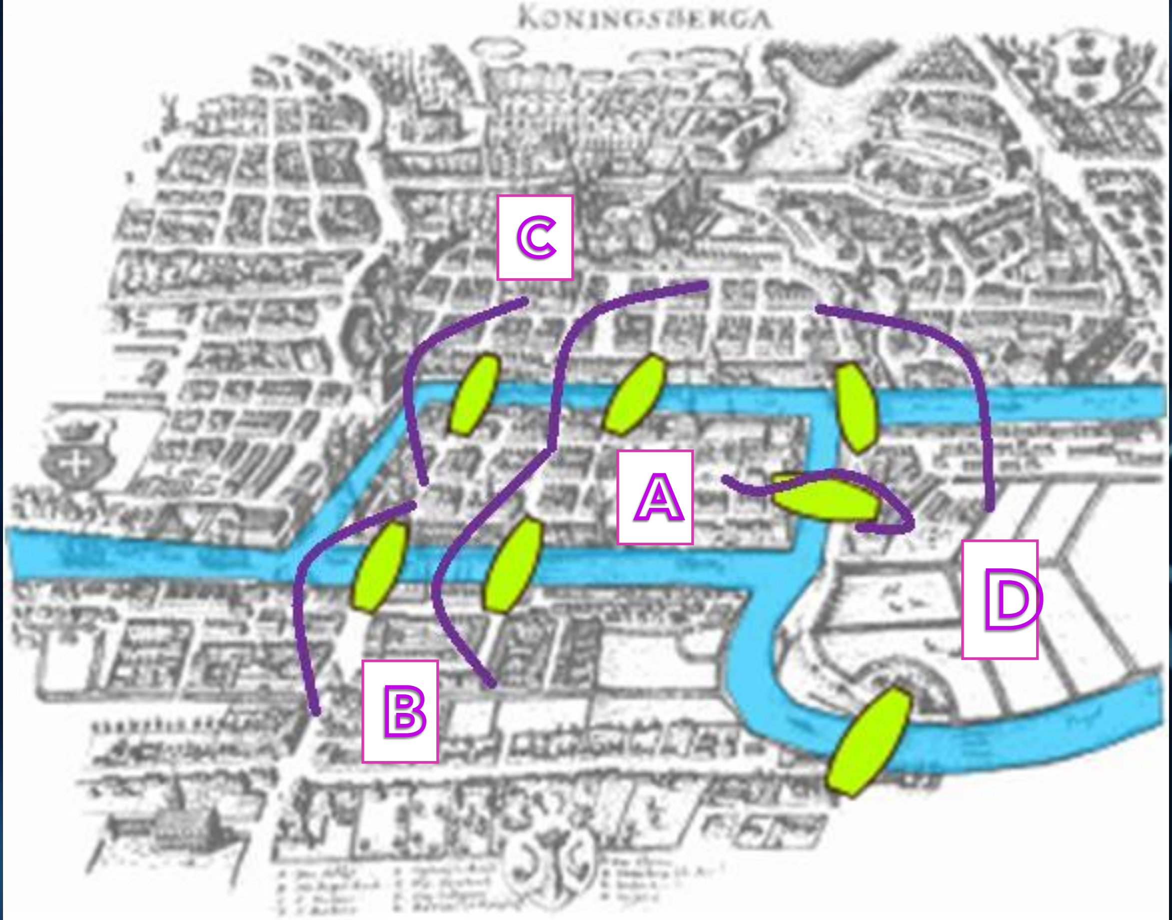
- Stories say that the people of Königsberg used to walk around the city on Sundays. While walking, they decided to create a game to play:
  - The initial puzzle requested a person to walk a path which crossed all bridges, crossing each bridge once and only once, and to finish at the starting point

The 7 bridges of Königsberg



- Stories say that the people of Königsberg used to walk around the city on Sundays. While walking, they decided to create a game to play:
  - The initial puzzle requested a person to walk a path which crossed all bridges, crossing each bridge once and only once, and to finish at the starting point
  - After many days and many trials, the puzzle was adjusted so that the finish point could be on any landmass, with the condition of crossing every bridge once and only once still holding

The 7 bridges of Königsberg



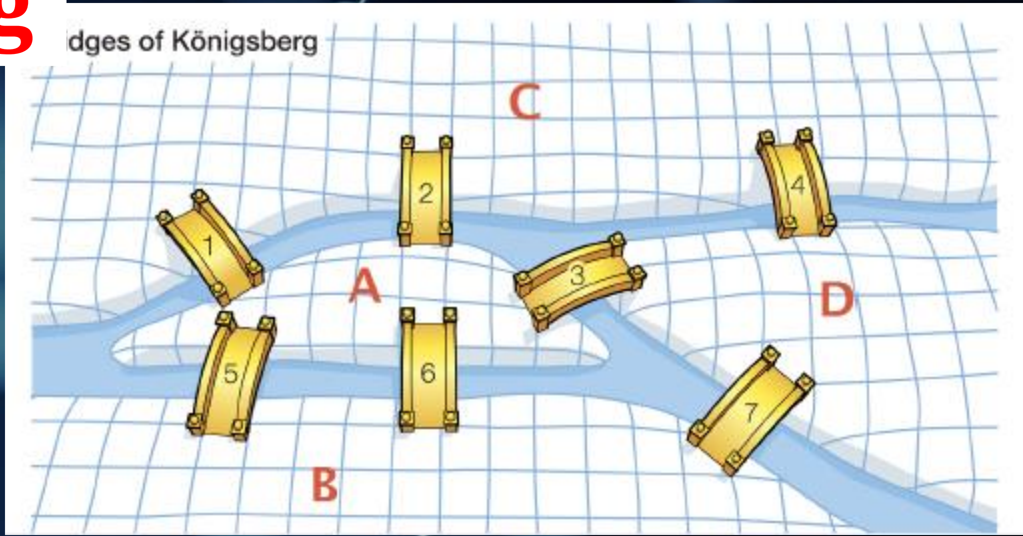
C->A->B->A->C->D->A???

4 land masses and 7 bridges

The question is whether it is possible to walk with a route that crosses each bridge exactly once?

The 7 bridges of Königsberg

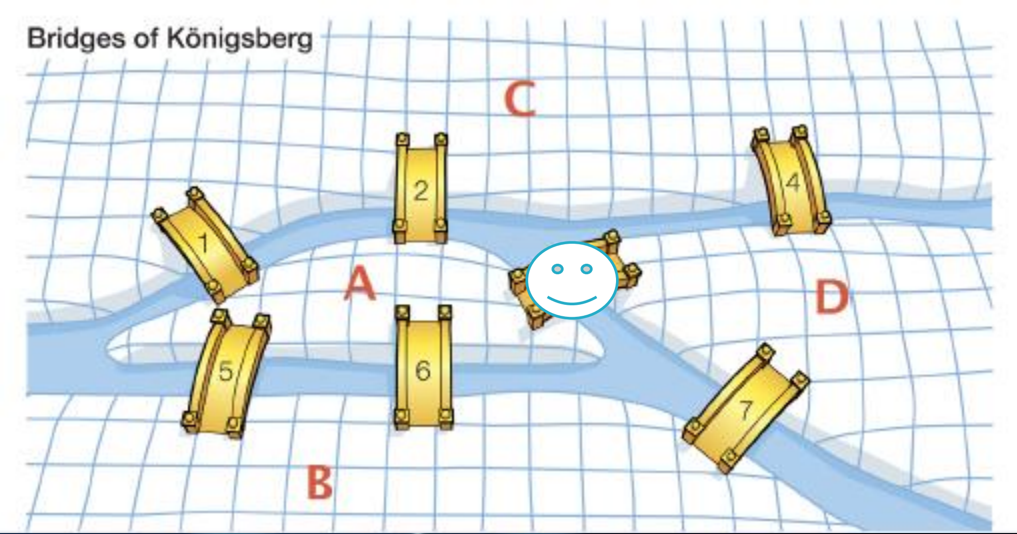
A-5 Edges; B-3 Edges
C-3 Edges; D-3 Edges



Euler gave the three guidelines that someone can use to figure out if a path exists using each bridge once and only once.

1. No journey is possible if there are **more than 2 odd landmasses**. This solves the Bridges of KönigsbergProblem since it contains 4 odd landmasses!
2. A journey is possible if there are **exactly two odd landmasses**. One can **start** in either one of the two odd landmasses and **end** on the other odd landmass.
3. A journey is possible if there are **no odd** landmasses. One starts and ends at any landmass in this case.

The 7 bridges of Königsberg



A-4 Edges; B-3 Edges
C-3 Edges; D-2 Edges

Guideline No:2

Euler gave the three guidelines that someone can use to figure out if a path exists using each bridge once and only once.

1. No journey is possible if there are **more than 2 odd landmasses**. This solves the Bridges of KönigsbergProblem since it contains 4 odd landmasses!
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3. A journey is possible if there are **no odd** landmasses. One starts and ends at any landmass in this case.

The Bridges Today

Today, the town of Königsberg is now part of East Prussia. It has a new name, Kaliningrad and the Pregel river was also renamed to Pregolya.

Figure: Königsberg (Kaliningrad) in modern day.

With only five bridges left, two nodes now have degree 2 and the other two have degree 3, so therefore, an Euclidian path is now possible, again. Although, it's impractical for tourists since it begins on one island and ends on the other.

Dynamical Processes in Complex Networks(SC1.440)

- Introduction to Computational Physics, Lecture of Prof. H. J. Herrmann
  Swiss Federal Institute of Technology ETH, Zürich, Switzerland
  Script by Dr. H. M. Singer, Lorenz Müller and Marco - Andrea Buchmann
  Computational Physics, IfB, ETH Zürich

- Nonlinear Dynamics –Steven Strogatz
- Networks: An Introduction –Newman
- Alain Barrat, , Marc Barthelemy, and Alessandro Vespignani. Dynamical processes on complex networks. Cambridge university press, 2008.

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- Boccaletti, Stefano, et al. “Complex networks: Structure and dynamics.”*Phyics reports, 424.4-5 (2006): 175-308.*
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- Perra, Nicola. "Non-pharmaceutical interventions during the COVID-19 pandemic: A review." *Physics Reports* 913 (2021): 1-52.

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- https://twitter.com/barabasi

Dynamical Processes in Complex Networks(SC1.440): Evaluations

Type of Evaluation	Weightage (in %)
Quiz (Jan end)	**20**
Mid Sem Exam (Feb end)	**30**
Quiz + Surprize quiz (after midsem)	**15+10**
Assignment plus viva /project	**25**

Dynamical Processes in Complex Networks(SC1.440): Class wise Plan

Class 1-2 (Jan 4,8)	**Introduction (Mod1), Networks metrices, Measures (Mod2)**	**Networks/bridge/papers / giants---**
Class 3 (Jan 11)	***Subrata Ghosh***	**Networkx and Gephi**
Class 4-5 (Jan 18,20)	**Networks metrices, Measures (Mod2) avg pathlength of chain lattice, EVC calculation, degree distribution**	**Demonstration with MatLab**
Class 6 (Jan 22)	**SIR model (deterministic), Networks and SIS/SIR Dynamics (Mod 3)- EVC and dynamics relation**	**Demonstration with MatLab**
Class 7 (Jan 25)	***Sasanka/Ujjwal (SIR simulation, SIR simulation in networks, test kit diffusion) Deterministic* (Mod 3)**	**Networkx and Gephi**

Class 9-10 (Feb 1, 5)	**Nonlinear Dynamics (Three bifurcations, Predator-prey, limit cycles, chaos) (Mod 3)**	**Demonstration with Matlab**
Class 11-13 (Feb 8, 15)	**Percolation, Erdos-Renyi Graphs (Mod 4a),**	**Demonstration with Matlab**
Class 13-14 (Feb 19, 22)	**ER-networks, Assignment/Project discussion for second half, Subrata**	**Networkx and Gephi Degree Distribution etc**
Midsem		**Exam**
Class 8 (Jan 29)	Quiz	

Dynamical Processes in Complex Networks(SC1.440): Classwise Plan

Class 15-16 (March 4,7)	Scale-free and small world(Mod 4b) (Aditi PGF)	Demonstration with MatLab
Class 17- (March 11)	Kuramoto dynamics, Chaotic dynamics, Networks (Mod5)	Demonstration with MatLab
Class 18 (March 14)	Mini quiz	
Class 19-20 (March 18)	Epidemics, Random walks, Ecology, Stability-Random Matrix	Demonstration with MatLab
Class 21 (March 21,28)	Project /Assignment discussion	

Class 22 (April 1) Quiz

Class 23-24 (April 4,8)	Subrata (PRL Vespignani, Hens Nature, Gao)	
(Class 25-28) April 15, 18, 22, 25	Project Viva	